

Algorithm Cheat Sheet — All Templates + Java Syntax

Conventions, Skeleton & Core API

Generic Test Harness

```
class Question {
    private int data;
    public Question(int data) {
        this.data = data;
    }
    public void solve() {
        // TODO
    }
}

public class Solution {
    public static void main(String[] args) {
        Question q = new Question(0);
        q.solve();
    }
}
```

Formatting Rules

```
if (condition) {
    doThis();
} else if (other) {
    doThat();
} else {
    fallback();
}
```

Naming Conventions (enforced across all template files)

Common Java API Cheat-Sheet

```
Map<Integer, Integer> map = new HashMap<>(); // key - value
Map<Integer, List<Integer>> adj = new HashMap<>(); // adjacency / buckets
Set<Integer> set = new HashSet<>(); // membership
List<Integer> list = new ArrayList<>(); // dynamic array
Deque<Integer> stack = new ArrayDeque<>(); // STACK: push / pop / peek
Deque<Integer> queue = new ArrayDeque<>(); // QUEUE: offer / poll / peek
// (ArrayDeque beats java.util.Stack / LinkedList)
PriorityQueue<Integer> minHeap = new PriorityQueue<>(); // min-heap (default)
PriorityQueue<Integer> maxHeap = new PriorityQueue<>(Collections.reverseOrder()); // max-heap
PriorityQueue<Integer> pq = new PriorityQueue<>((a, b) -> a[1] - b[1]); // by field, min-first
TreeMap<Integer, Integer> tmap = new TreeMap<>(); // sorted keys: floor/ceiling/first/last
TreeSet<Integer> tset = new TreeSet<>(); // sorted set: floor/ceiling/higher/lower
LinkedHashSet<Integer> lset = new LinkedHashSet<>(); // insertion-ordered set (LIFO buckets)
Random rand = new Random(); // rand.nextInt(n) -> 0..n-1 (RandomizedSet)
int[] arr = new int[n]; // defaults to 0
boolean[] seen = new boolean[n]; // defaults to false
int[][] grid = new int[m][n]; // 2D, all 0
int[] dr = {1,-1,0,0}, dc = {0,0,1,-1}; // DOWN,UP,RIGHT,LEFT;
iterate for(dir=0..3) {
    idr[dir],j+dc[dir]
}
```

```
// — Map —
map.put(k, v);
map.getOrDefault(k, 0); // safe read with default
map.merge(k, 1, Integer::sum); // + counting
map.computeIfAbsent(k, x -> new ArrayList<>()).add(v); // + build adjacency / buckets
map.putIfAbsent(k, v); // - write only first occurrence (keep earliest index)
map.containsKey(k); map.remove(k);
for (Map.Entry<Integer,Integer> e : map.entrySet()) { e.getKey(); e.getValue(); e.setValue(v); }
map.keySet(); map.values();
// — Set —
set.add(x); set.contains(x); set.remove(x);
// — List —
list.add(x); list.get(i); list.set(i, x); // - pop in
list.remove(list.size() - 1); // - backtracking
list.size(); list.isEmpty(); list.addAll(other);
Collections.sort(list); // ascending
list.sort((a, b) -> b - a); // custom / descending
Collections.reverse(list); // - reverse path
after building it backwards
```

```
// Deque as STACK (LIFO) // Deque as QUEUE (FIFO)
stack.push(x); queue.offer(x);
int top = stack.pop(); int head = queue.poll();
int look = stack.peek(); int look = queue.peek();
stack.isEmpty(); queue.isEmpty();
// Deque BOTH ENDS - needed for the MONOTONE DEQUE (sliding-window max/min)
// and for addFirst-style level building (zigzag BFS).
deque.offerLast(x); deque.pollLast(); deque.peekLast(); // back (push/pop/peek tail)
deque.offerFirst(x); deque.pollFirst(); deque.peekFirst(); // front (push/pop/peek head)
deque.addLast(x); deque.addFirst(x); // same as offer*, throw if capacity-bound
// PriorityQueue (heap)
pq.offer(x); int best = pq.poll(); int look = pq.peek(); pq.size();
pq.remove(x); // - O(n) removal of an arbitrary element (lazy-deletion alternative in #480)
```

```
Integer.MAX_VALUE; // 2147483647 (2^31 - 1) - sentinel for "min so far"
Integer.MIN_VALUE; // -2147483648 (-2^31) - sentinel for "max so far"
Long.MAX_VALUE; // when sums can overflow int, accumulate in long
int k = i + (j - i) / 2; // - avoid (i + j) overflow in binary search
Integer.compare(a, b); // - overflow-safe comparator; use instead of (a - b)
(a, b) -> Integer.compare(a[1], b[1]); // - safe PriorityQueue/Arrays.sort comparator
Double.compare(a, b); // - comparator for double[] heaps (probability, median)
```

```
ch - 'a'; // 'a'..'z' -> 0..25 (lowercase)
bucket index)
ch - '0'; // '0'..'9' -> 0..9 (digit value)
(char) ('a' + i); // 0..25 -> 'a'..'z'
num = num * 10 + (ch - '0'); // - build a number digit by digit
Integer.parseInt("123"); // String -> int (throws on bad input)
Integer.valueOf(123); // int -> Integer (boxed)
String.valueOf(123); // int/char/etc -> String
Character.getNumericValue('7'); // char -> 7
// char classification
Character.isDigit(ch); Character.isLetter(ch);
Character.isLetterOrDigit(ch);
Character.isWhitespace(ch); Character.toLowerCase(ch);
Character.toUpperCase(ch);
```

```
s.length(); s.charAt(i); s.substring(i, j); // [i, j)
s.toCharArray(); s.split(" "); s.equals(t); s.compareTo(t);
s.indexOf("ab"); s.isEmpty();
s.startsWith("ab"); s.endsWith("z"); s.contains("x"); // prefix / suffix / substring tests
s.toLowerCase(); // case-normalize (e.g. case-insensitive compare)
s.trim(); s.replace('a', 'b'); s.repeat(n); // strip ends / swap chars / repeat
StringBuilder builder = new StringBuilder();
builder.append(x); builder.reverse(); builder.toString();
builder.charAt(i); builder.setCharAt(i, c); builder.deleteCharAt(i);
builder.length();
String.join(",", list); // list -> "a,b,c"
```

```
Arrays.sort(arr); // ascending in place
Arrays.sort(intervals, (a, b) -> a[0] - b[0]); // 2D by first field
Arrays.fill(dp, Integer.MAX_VALUE); // seed a dp/dist array
Arrays.equals(window, need); // - element-wise array compare (anagram check)
Arrays.copyOf(arr, len); Arrays.copyOfRange(arr, i, j);
System.arraycopy(src, srcPos, dst, dstPos, len); // - fast block copy (merge step)
Arrays.asList(1, 2, 3); // fixed-size List
view
int sum = Arrays.stream(arr).sum();
int max = Arrays.stream(arr).max().getAsInt();
// List<Integer> -> int[]
int[] a = List.stream(list).mapToInt(Integer::intValue).toArray();
List<Integer> l = Arrays.stream(a).boxed().collect(Collectors.toList());
```

```
Math.max(a, b); Math.min(a, b); Math.abs(x);
Math.pow(a, b); Math.sqrt(x); Math.ceil(x); Math.floor(x);
n & (n - 1); // clear lowest set bit
n & (-n); // isolate lowest set bit
1 << i; // bit i set (use 1L << i for i ≥ 31)
(n >> i) & 1; // read bit i
(n & 1) == 0; // - even (last bit 0)
(n & 1) == 1; // - odd (last bit 1)
n >> 1; // - divide by 2 (drop last bit); n << 1 = multiply by 2
Integer.bitCount(n); // # of set bits
Integer.toBinaryString(n);
```

```
tmap.firstKey(); tmap.lastKey();
tmap.floorKey(x); // largest key ≤ x tmap.ceilingKey(x); // smallest key ≥ x
tmap.lowerKey(x); // largest key < x tmap.higherKey(x); // smallest key > x
tset.floor(x); tset.ceiling(x); tset.lower(x); tset.higher(x);
tset.first(); tset.last();
```

```
// - NEGATIVE-SAFE MODULO - replaces the manual ((x % n) + n) % n
Math.floorMod(x, n); // always in [0, n-1] even when x < 0
(prefix-sum % k)
int[] int[] INSTEAD OF HashMap for a bounded key space —
int[] count = new int[26]; // O(1) no-boxing freq vs
HashMap<Character,Integer> // also faster to compare:
count[ch - 'a']++;
Arrays.equals(a, b) // 1D ROLLING ARRAY for DP — drop a dimension when row i only needs row i-1
int[] dp = new int[n + 1]; // O(n) space instead of O(m*n)
for (...) for (int j = target; j >= w; j--) dp[j] += dp[j - w]; // 0/1 knapsack: iterate j DOWN
// - FIXED-SIZE HEAP for top-k — O(n log k) beats sorting O(n log n)
if (heap.size() > k) heap.poll(); // keep only k best; min-heap for "k largest"
// - LAZY DELETION over pq.remove(obj) — pq.remove is O(n); skip stale entries instead
if (entry[0] != dist[node]) continue; // Dijkstra: ignore outdated heap entries
// - ArrayDeque over Stack / LinkedList — faster, no synchronization, both ends O(1)
Deque<Integer> stack = new ArrayDeque<>(); // never use java.util.Stack
// — Stringing/Deque over String — in a loop — O(1)
StringBuilder builder = new StringBuilder(); // append, then builder.toString() once
// — EARLY EXIT / PRUNING — return the moment the answer is decided
if (found) return result; // e.g. Trie shortest-root, backtracking bounds
```

Template Index

Two Pointers

Pattern 1 — Opposite Two Pointers

```
int i = 0, j = n - 1;
while (i < j) {
    if (condition == target) {
        // record #/ i++; j--;
    } else if (tooSmall) {
        i++;
    } else {
        j--;
    }
}
```

Sliding Window

The Three Templates

```
int i = 0;
for (int j = 0; j < n; j++) {
    if (j - i + 1 == k) {
        i++;
    }
}
int i = 0, result = 0;
for (int j = 0; j < n; j++) {
    while (/* window exceeds constraint */) {
        i++;
    }
    result = Math.max(result, j - i + 1);
}
int i = 0, result = Integer.MAX_VALUE;
for (int j = 0; j < n; j++) {
    while (/* window VALID */) {
        result = Math.min(result, j - i + 1);
        i++;
    }
}
```

Frequency Map Trick (used by 567 and 76)

```
if (need[s.charAt(j)]-- > 0) required--;
if (need[s.charAt(i)]++ >= 0) required++;
i++;
```

Prefix Sum + HashMap

Core Template

```
Map<Long, Integer> map = new HashMap<>();
map.put(0L, ???);
long sum = 0;
for (int i = 0; i < nums.length; i++) {
    sum += nums[i];
    long lookup = transform(sum);
    map.put(storeKey(sum), storeValue(i)); // what to store (sum or sum%k - count or index)
}
```

Binary Search

The Only Template — [i, j), find first k with condition(k)

```
int i = 0, j = nums.length;
while (i < j) {
    int k = i + (j - i) / 2;
    if (condition(k)) {
        i = k + 1;
    } else {
        j = k;
    }
}
```

How to Choose

Sorting

Template 1 — Merge Sort

```
public void mergeSort(int[] nums) {
    mergeSort(nums, 0, nums.length, new int[nums.length]);
}
private void mergeSort(int[] nums, int start, int end, int[] temp) {
    if (end - start <= 1) return;
    int mid = start + (end - start) / 2;
    mergeSort(nums, start, mid, temp);
    mergeSort(nums, mid, end, temp);
    merge(nums, start, mid, end, temp);
}
private void merge(int[] nums, int start, int mid, int end, int[] temp) {
    int i = start, j = mid, k = start;
    while (i < mid || j < end) {
        if (j >= end || (i < mid && nums[i] <= nums[j])) {
            temp[k++] = nums[i++];
        } else {
            temp[k++] = nums[j++];
        }
    }
    for (i = start; i < end; i++) nums[i] = temp[i];
}
```

Template 2 — Quick Sort (3-way Dutch Flag Partition)

```
public void quickSort(int[] nums) {
    quickSort(nums, 0, nums.length);
}
private void quickSort(int[] nums, int start, int end) {
    if (end - start <= 1) return;
    int pivot = nums[start + (end - start) / 2];
    int i = start, k = start, j = end - 1;
    while (k <= j) {
        if (nums[k] < pivot) {
            swap(nums, k++, i++);
        } else if (nums[k] == pivot) {
            k++;
        } else {
            swap(nums, k, j--);
        }
    }
    quickSort(nums, start, i);
    quickSort(nums, k, end);
}
private void swap(int[] nums, int i, int j) {
    int t = nums[i]; nums[i] = nums[j]; nums[j] = t;
}
```

Template 3 — Quick Select (partial sort — k-th element)

```
private int quickSelect(int[] nums, int start, int end, int target) {
    int pivot = nums[start + (end - start) / 2];
    int i = start, k = start, j = end - 1;
    while (k <= j) {
        if (nums[k] < pivot) {
            swap(nums, k++, i++);
        } else if (nums[k] == pivot) {
            k++;
        } else {
            swap(nums, k, j--);
        }
    }
    if (target < i) {
        return quickSelect(nums, start, i, target);
    } else if (target < k) {
        return pivot;
    } else {
        return quickSelect(nums, k, end, target);
    }
}
```

Template 4 — Counting Sort

```
public void countingSort(int[] arr) {
    if (arr.length == 0) return;
    int min = Arrays.stream(arr).min().getAsInt();
    int max = Arrays.stream(arr).max().getAsInt();
    int[] buckets = new int[max - min + 1];
    for (int x : arr) buckets[x - min]++;
    int i = 0;
    for (int v = 0; v < buckets.length; v++)
        while (buckets[v]-- > 0) arr[i++] = v + min;
}
```

Linked List

When to Use — Signal → Pattern

All Four Templates

```
ListNode sentinel = new ListNode(0);
sentinel.next = head;
ListNode previous = sentinel, current = head;
while (current != null) {
    if (shouldDelete(current)) {
        previous.next = current.next;
    } else {
        previous = current;
    }
    current = current.next;
}
return sentinel.next;
ListNode previous = null, current = head;
while (current != null) {
    ListNode next = current.next;
    current.next = previous;
    previous = current;
    current = next;
}
return previous;
ListNode slow = head, fast = head;
while (fast != null && fast.next != null) {
    slow = slow.next;
    fast = fast.next.next;
}
ListNode sentinel = new ListNode(0);
ListNode current = sentinel;
while (a != null && b != null) {
    if (a.val <= b.val) {
        current.next = a;
        a = a.next;
    } else {
        current.next = b;
        b = b.next;
    }
    current = current.next;
}
current.next = (a != null) ? a : b;
return sentinel.next;
```

String

Core Idioms

```
int[] count = new int[26];
for (char ch : s.toCharArray()) count[ch - 'a']++;
int[] count = new int[10];
for (char ch : s.toCharArray()) count[ch - '0']++;
int num = 0;
for (int i = 0; i < s.length(); i++)
    num = num * 10 + (s.charAt(i) - '0');
String hashCode(String s) {
    int[] count = new int[26];
    for (char ch : s.toCharArray()) count[ch - 'a']++;
    StringBuilder builder = new StringBuilder();
    for (int i = 0; i < 26; i++) {
        builder.append((char)('a' + i));
        builder.append(count[i]);
    }
    return builder.toString();
}
char[] chars = s.toCharArray();
Arrays.sort(chars);
String key = new String(chars);
int[] count = new int[128];
for (char ch : s.toCharArray()) count[ch]++;
List<Character>[] buckets = new List[s.length() + 1]; // index = frequency
for (int i = 0; i < 128; i++) {
    if (count[i] > 0) {
        int freq = count[i];
        if (buckets[freq] == null) buckets[freq] = new ArrayList<>();
        buckets[freq].add((char) i);
    }
}
StringBuilder builder = new StringBuilder();
for (int freq = buckets.length - 1; freq > 0; freq--) {
    if (buckets[freq] == null) continue;
    for (char ch : buckets[freq])
        for (int i = 0; i < freq; i++) builder.append(ch);
}
int expand(String s, int i, int j) {
    while (i >= 0 && j < s.length() && s.charAt(i) == s.charAt(j)) { i--; j++; }
    return j - i - 1;
}
```

Stack / Queue / Heap

All Templates

```
Deque<Integer> stack = new ArrayDeque<>();
stack.push(x);
stack.pop();
stack.peek();
Deque<Integer> stack = new ArrayDeque<>();
for (int i = 0; i < n; i++) {
    while (!stack.isEmpty() && nums[stack.peek()] < nums[i])
        result[stack.pop()] = nums[i];
    stack.push(i);
}
Deque<Integer> stack = new ArrayDeque<>();
for (int i = 0; i < n; i++) {
    while (!stack.isEmpty() && nums[stack.peek()] > nums[i]) {
        int height = nums[stack.pop()];
        int width = stack.isEmpty() ? i : i - stack.peek() - 1;
    }
    stack.push(i);
}
Deque<Integer> queue = new ArrayDeque<>();
for (int i = 0; i < n; i++) {
    while (!queue.isEmpty() && nums[queue.peekLast()] <= nums[i])
        queue.pollLast();
    queue.offerLast(i);
    if (queue.peekFirst() < i - k + 1) queue.pollFirst();
    if (i >= k - 1) result[i - k + 1] = nums[queue.peekFirst()];
}
PriorityQueue<Integer> minPQ = new PriorityQueue<>();
PriorityQueue<Integer> maxPQ = new PriorityQueue<>()
    (Collections.reverseOrder()); // max-heap
PriorityQueue<int[]> pq = new PriorityQueue<>((a, b) -> a[1] - b[1]);
// custom: sort by index 1
pq.offer(x);
pq.poll();
pq.peek();
PriorityQueue<Integer> maxHeap = new PriorityQueue<>()
    (Collections.reverseOrder());
PriorityQueue<Integer> minHeap = new PriorityQueue<>();
void addNum(int num) {
    maxHeap.offer(num);
    minHeap.offer(maxHeap.poll());
    if (maxHeap.size() < minHeap.size())
        maxHeap.offer(minHeap.poll());
}
double findMedian() {
    return maxHeap.size() > minHeap.size()
        ? maxHeap.peek()
        : (maxHeap.peek() + minHeap.peek()) / 2.0;
}
```

BFS (Tree)

The Template — Level-Aware BFS

```
Queue<TreeNode> queue = new ArrayDeque<>();
queue.offer(root);
int level = 0;
while (!queue.isEmpty()) {
    int size = queue.size();
    level++;
    for (int i = 0; i < size; i++) {
        TreeNode node = queue.poll();
        if (node.left != null) queue.offer(node.left);
        if (node.right != null) queue.offer(node.right);
    }
}
```

When to Use — Signal → Pattern

The One Rule — Snapshot Level Size First

Always snapshot the level size BEFORE the inner `for` loop:
for (int i = 0; i < size; i++) { ... }
Without the snapshot, newly added children mix with the current level and `i = size-1 / i = 0` checks become meaningless.

DFS / Backtracking

When to Use — Signal → Pattern

All Templates

```
private int dfs(TreeNode node) {
    if (node == null) return BASE;
    int left = dfs(node.left);
    int right = dfs(node.right);
    return ...
}
private void dfs(TreeNode node, int accumulated) {
    if (node == null) return;
    accumulated += node.val;
    if (isLeaf(node)) { /* record */ return; }
    dfs(node.left, accumulated);
    dfs(node.right, accumulated);
}
int answer = Integer.MIN_VALUE;
private int dfs(TreeNode node) {
    if (node == null) return 0;
    int left = Math.max(0, dfs(node.left));
    int right = Math.max(0, dfs(node.right));
    answer = Math.max(answer, left + right + node.val);
    return Math.max(left, right) + node.val;
}
private void dfs(int[][] grid, int r, int c) {
    if (r < 0 || r >= grid.length || c < 0 || c >= grid[0].length) return;
    if (grid[r][c] != TARGET) return;
    grid[r][c] = VISITED;
    dfs(grid, r+1, c); dfs(grid, r-1, c);
    dfs(grid, r, c+1); dfs(grid, r, c-1);
}
int[] state;
private boolean dfs(int node) {
    if (state[node] == 1) return true;
    if (state[node] == 2) return false;
    state[node] = 1;
    for (int neighbor : graph.get(node))
        if (dfs(neighbor)) return true;
    state[node] = 2;
    return false;
}
private void backtrack(int start, List<Integer> current) {
    if (baseCase) { result.add(new ArrayList<>(current)); return; }
    for (int i = start; i < n; i++) {
        if (skipCondition(i)) continue;
        current.add(nums[i]);
        backtrack(next, current);
        current.remove(current.size()-1);
    }
}
```

Graph

Complexity Legend

BFS / DFS	$O(V + E)$	visit each vertex and edge once
Topo sort (Kahn)	$O(V + E)$	each node enqueued once, each edge relaxed once
Union-Find	$\sim O(n \cdot \alpha(n)) = O(n)$	α = inverse Ackermann, effectively constant
Dijkstra	$O((V + E) \log V)$	non-negative weights ONLY

Graph Representation

```
List<List<Integer>> graph = new ArrayList<>();
for (int i = 0; i < n; i++) graph.add(new ArrayList<>());
for (int[] edge : edges) {
    graph.get(edge[0]).add(edge[1]);
    graph.get(edge[1]).add(edge[0]);
}
List<List<int[]>> graph = new ArrayList<>();
for (int i = 0; i < n; i++) graph.add(new ArrayList<>());
for (int[] edge : edges)
    graph.get(edge[0]).add(new int[]{edge[1], edge[2]});
int[] dr = {1, -1, 0, 0};
int[] dc = {0, 0, 1, -1};
```

Core Templates

```
Queue<Integer> queue = new ArrayDeque<>();
boolean[] visited = new boolean[n];
queue.offer(start);
visited[start] = true;
int level = 0;
while (!queue.isEmpty()) {
    int size = queue.size();
    for (int i = 0; i < size; i++) {
        int node = queue.poll();
        for (int neighbor : graph.get(node)) {
            if (!visited[neighbor]) {
                visited[neighbor] = true;
                queue.offer(neighbor);
            }
        }
    }
    level++;
}
int[] inDegree = new int[n];
for (int[] edge : edges) inDegree[edge[1]]++;
Queue<Integer> queue = new ArrayDeque<>();
for (int i = 0; i < n; i++) if (inDegree[i] == 0) queue.offer(i);
List<Integer> order = new ArrayList<>();
while (!queue.isEmpty()) {
    int node = queue.poll();
    order.add(node);
    for (int neighbor : graph.get(node))
        if (--inDegree[neighbor] == 0) queue.offer(neighbor);
}
int[] parent, rank;
void init(int n) {
    parent = new int[n]; rank = new int[n];
    for (int i = 0; i < n; i++) parent[i] = i;
}
int find(int x) {
    if (parent[x] != x) parent[x] = find(parent[x]);
    return parent[x];
}
boolean union(int x, int y) {
    int px = find(x), py = find(y);
    if (px == py) return false;
    if (rank[px] < rank[py]) { int t = px; px = py; py = t; }
    parent[py] = px;
    if (rank[px] == rank[py]) rank[px]++;
    return true;
}
int[] dist = new int[n];
Arrays.fill(dist, Integer.MAX_VALUE);
dist[src] = 0;
PriorityQueue<int[]> pq = new PriorityQueue<>((a, b) -> a[1] - b[1]);
pq.offer(new int[]{src, 0});
while (!pq.isEmpty()) {
    int[] current = pq.poll();
    int node = current[0], d = current[1];
    if (d > dist[node]) continue;
    for (int[] nb : graph.get(node)) {
        int next = nb[0], w = nb[1];
        if (dist[node] + w < dist[next]) {
            dist[next] = dist[node] + w;
            pq.offer(new int[]{next, dist[next]});
        }
    }
}
int[] color = new int[n];
Arrays.fill(color, -1);
Queue<Integer> queue = new ArrayDeque<>();
for (int start = 0; start < n; start++) {
    if (color[start] == -1) continue;
    color[start] = 0;
    queue.offer(start);
    while (!queue.isEmpty()) {
        int node = queue.poll();
        for (int neighbor : graph.get(node)) {
            if (color[neighbor] == -1) { color[neighbor] = 1 - color[node]; queue.offer(neighbor); }
            else if (color[neighbor] == color[node]) return false;
        }
    }
}
return true;
```

Greedy

Two Interval Templates

MERGE DIRECTION RULE (for merge-style problems like #56, where both keys work):
Sort by START → traverse FORWARD ($i = 0 \rightarrow n-1$), extend the END of last kept
Sort by END → traverse BACKWARD ($i = n-1 \rightarrow 0$), extend the START of last kept
These two are exact mirror images. See #56 for both written out side by side.
(NOTE: this mirror only applies to merging. Template 1 below sorts by end but still sweeps FORWARD – there the goal is counting, not merging.)
TEMPLATE 1 – Sort by END time, sweep forward, track end of last kept interval
Use when: maximizing count of non-overlapping intervals
Greedy insight: earliest-finishing interval leaves most room for future ones
TEMPLATE 2 – Sort by START time, sweep forward, merge when overlapping
Use when: merging/counting overlapping intervals
Greedy insight: processing in start order → each interval only needs to compare with the last merged result

```
Arrays.sort(intervals, (a, b) -> a[1] - b[1]);
int lastEnd = Integer.MIN_VALUE;
for (int[] interval : intervals) {
    if (interval[0] >= lastEnd) {
        lastEnd = interval[1];
    } else {
    }
}
Arrays.sort(intervals, (a, b) -> a[0] - b[0]);
List<int[]> result = new ArrayList<>();
for (int[] interval : intervals) {
    if (result.isEmpty() || result.get(result.size()-1)[1] < interval[0]) {
        result.add(interval);
    } else {
        result.get(result.size()-1)[1] =
            Math.max(result.get(result.size()-1)[1], interval[1]);
    }
}
Arrays.sort(intervals, (a, b) -> a[0] - b[0]);
PriorityQueue<Integer> pq = new PriorityQueue<>();
for (int[] interval : intervals) {
    if (!pq.isEmpty() && pq.peek() <= interval[0]) {
        pq.poll();
    }
    pq.offer(interval[1]);
}
```

Dynamic Programming

All Six Templates

```
int[] dp = new int[n + 1];
dp[0] = base;
for (int i = 1; i <= n; i++) {
    dp[i] = f(dp[i-1], dp[i-2], ...);
}
int[] dp = new int[n];
for (int i = 0; i < n; i++) {
    for (int j = 0; j < i; j++) {
        dp[i] = f(dp[i], dp[j]);
    }
}
```

Want count or min/max?

- Count (dp[j] += dp[j - num])
 - 0/1 (each item once): j descending → #416, #494
 - Unbounded (reuse): j ascending → #518
 - Ordered (perms): target outer, nums inner → #377
- Min/Max (dp[j] = min/max(...))
 - 0/1 minimize: j descending → #474 (maximize)
 - Unbounded minimize: j ascending → #322

```
int[] dp = new int[target + 1];
dp[0] = 1;
for (int i = 0; i < nums.length; i++) {
    for (int j = target; j >= nums[i]; j--) {
        dp[j] += dp[j - nums[i]];
    }
}
int[] dp = new int[target + 1];
dp[0] = 1;
for (int i = 0; i < nums.length; i++) {
    for (int j = nums[i]; j <= target; j++) {
        dp[j] += dp[j - nums[i]];
    }
}
int[] dp = new int[target + 1];
dp[0] = 1;
for (int j = 1; j <= target; j++) {
    for (int num : nums) {
        if (j >= num) dp[j] += dp[j - num];
    }
}
```

```
int[][] dp = new int[m + 1][n + 1];
for (int i = 1; i <= m; i++) {
    for (int j = 1; j <= n; j++) {
        if (s1.charAt(i-1) == s2.charAt(j-1)) {
            dp[i][j] = dp[i-1][j-1] + 1;
        } else {
            dp[i][j] = combine(dp[i-1][j], dp[i][j-1], dp[i-1][j-1]);
        }
    }
}
```

```
int[][] dp = new int[n][n];
for (int i = 0; i < n; i++) dp[i][i] = base_case;
for (int i = n - 1; i >= 0; i--) {
    for (int j = i + 1; j < n; j++) {
        dp[i][j] = ...;
    }
}
```

```
int[][] dp = new int[n][n];
for (int i = 0; i < n; i++) dp[i][i] = base_case;
for (int i = 0; i < n; i++) {
    for (int j = i - 1; j >= 0; j--) {
        dp[i][j] = ...;
    }
}
```

```
int[][] dp = new int[n][n];
for (int i = 0; i < n; i++) dp[i][0] = 1;
for (int i = 1; i < n; i++) {
    for (int j = 1; j <= i; j++) {
        dp[i][j] = ...;
    }
}
```

```
int[] dr = {1,-1,0,0}, dc = {0,0,1,-1};
int[][] dp = new int[rows][cols];
for (int i = 0; i < rows; i++) {
    for (int j = 0; j < cols; j++) {
        dp[i][j] = grid[i][j] + f(dp[i-1][j], dp[i][j-1]);
    }
}
int[][] memo = new int[rows][cols];
for (int i = 0; i < rows; i++) {
    for (int j = 0; j < cols; j++) {
        dfs(grid, i, j, memo, dr, dc);
    }
}
```

cash = max profit when NOT holding (free to buy or idle)
hold = max profit when HOLDING stock (bought it at some cost)
cooldown = max profit right after selling (can't buy today)

	buy	sell	re-buy
condition			
#121 (1 tx): (ignore cash)	hold = max(hold, -price)		no re-buy
#122 (unlimited): full cash	hold = max(hold, cash - price)		re-buy with
#123 (2 tx): chained	chain: buy2 uses sell1 cash		4 states
#309 (cooldown):	hold = max(hold, cooldown - price)		must wait 1 day
#714 (fee):	hold = max(hold, cash - price)		pay fee on sell

Trie

When to Use Trie (vs HashMap/Array)

Canonical Template

```
class TrieNode {
    TrieNode[] children = new TrieNode[26];
    boolean isEnd = false;
}
class TrieNode {
    Map<Character, TrieNode> children = new HashMap<>();
    boolean isEnd = false;
}
void insert(TrieNode root, String word) {
    TrieNode node = root;
    for (char c : word.toCharArray()) {
        int idx = c - 'a';
        if (node.children[idx] == null) node.children[idx] = new
        TrieNode();
        node = node.children[idx];
    }
    node.isEnd = true;
}
boolean search(TrieNode root, String word) {
    TrieNode node = root;
    for (char c : word.toCharArray()) {
        int idx = c - 'a';
        if (node.children[idx] == null) return false;
        node = node.children[idx];
    }
    return node.isEnd;
}
boolean startsWith(TrieNode root, String prefix) {
    TrieNode node = root;
    for (char c : prefix.toCharArray()) {
        int idx = c - 'a';
        if (node.children[idx] == null) return false;
        node = node.children[idx];
    }
    return true;
}
```

Variations

```
boolean dfs(TrieNode node, String word, int i) {
    if (i == word.length()) return node.isEnd;
    char c = word.charAt(i);
    if (c == '.') {
        for (TrieNode child : node.children) {
            if (child != null && dfs(child, word, i+1)) return true;
        }
    }
    int idx = c - 'a';
    if (node.children[idx] == null) return false;
    return dfs(node.children[idx], word, i + 1);
}
String findRoot(TrieNode root, String word) {
    TrieNode node = root;
    for (int i = 0; i < word.length(); i++) {
        int idx = word.charAt(i) - 'a';
        if (node.children[idx] == null) return word;
        node = node.children[idx];
        if (node.isEnd) return word.substring(0, i+1);
    }
    return word;
}
void insert(TrieNode root, String word) {
    TrieNode node = root;
    for (char c : word.toCharArray()) {
        int idx = c - 'a';
        if (node.children[idx] == null) node.children[idx] = new
        TrieNode();
        node = node.children[idx];
        if (node.words.size() < 3) node.words.add(word);
    }
    node.isEnd = true;
}
void dfs(char[][] board, int i, int j, TrieNode node, StringBuilder path,
Set<String> result) {
    if (i < 0 || i >= board.length || j < 0 || j >= board[0].length)
    return;
    char c = board[i][j];
    if (c == '#' || node.children[c - 'a'] == null) return;
    node = node.children[c - 'a'];
    path.append(c);
    if (node.isEnd) result.add(path.toString());
    board[i][j] = '#';
    dfs(board, i+1, j, node, path, result);
    dfs(board, i-1, j, node, path, result);
    dfs(board, i, j+1, node, path, result);
    dfs(board, i, j-1, node, path, result);
    board[i][j] = c;
    path.deleteCharAt(path.length() - 1);
}
```

search vs startsWith — One Line Difference

```
search:      return node.isEnd;
startsWith:  return true;
```

Bit Manipulation

Canonical Template

```
int result = 0;
for (int num : nums) result ^= num;
boolean isSet = ((n >> 1) & 1) == 1;
int set = n | (1 << i);
int clear = n & ~(1 << i);
int toggle = n ^ (1 << i);
int lowest = n & (-n);
boolean isPow2 = n > 0 && (n & (n-1)) == 0;
int count = 0;
while (n != 0) { count++; n &= (n - 1); }
int mask = 0;
for (char c : word.toCharArray()) mask |= (1 << (c - 'a'));
if ((mask[i] & mask[j]) == 0) { /* no shared letter */ }
```

Variations

```
int ones = 0, twos = 0;
for (int num : nums) {
    ones = (ones ^ num) & ~twos;
    twos = (twos ^ num) & ~ones;
}
int xor = 0;
for (int num : nums) xor ^= num;
int diff = xor & (-xor);
int a = 0;
for (int num : nums) {
    if ((num & diff) != 0) a ^= num;
}
int b = xor ^ a;
int[] dp = new int[n + 1];
for (int i = 1; i <= n; i++) {
    dp[i] = dp[i >> 1] + (i & 1);
    int shift = 0;
    while (left != right) { left >>= 1; right >>= 1; shift++; }
    return left << shift;
}
int add(int a, int b) {
    while (b != 0) {
        int carry = a & b;
        a = a ^ b;
        b = carry << 1;
    }
    return a;
}
```

Design

Canonical Template — HashMap + Doubly Linked List (LRU)

```
class CacheNode {
    int key, value;
    CacheNode previous, next;
    CacheNode(int key, int value) { this.key = key; this.value = value; }
}
CacheNode head = new CacheNode(0, 0), tail = new CacheNode(0, 0);
head.next = tail; tail.previous = head;
void remove(CacheNode node) {
    node.previous.next = node.next;
    node.next.previous = node.previous;
}
void insertAfterHead(CacheNode node) {
    node.next = head.next;
    node.previous = head;
    head.next.previous = node;
    head.next = node;
}
CacheNode lru = tail.previous;
remove(lru);
map.remove(lru.key);
```

Variations

Math

Canonical Templates

```
while (n != 0) {
    int digit = n % 10;
    n /= 10;
}
double fastPow(double x, long n) {
    double result = 1;
    while (n > 0) {
        if ((n & 1) == 1) result *= x;
        x *= x;
        n >>= 1;
    }
    return result;
}
StringBuilder builder = new StringBuilder();
while (num > 0) {
    builder.append(num % b);
    num /= b;
}
builder.reverse();
int value = 0;
for (int digit : digits) value = value * b + digit;
boolean[] composite = new boolean[n];
for (int i = 2; i * i <= n; i++) {
    if (composite[i]) continue;
    for (int j = i * i; j <= n; j += i) {
        composite[j] = true;
    }
}
```